Subject Name: Generative AI with Vector Databases
Subject Code: CSU 2217
Total Marks: 30
Exam: Term 2
Set E
B.Tech 2nd Year
Group – 1
Total Number = 31

Instructions:
The Exam is open book – You are allowed to bring books, lecture notes, slides.
No electronic gadgets are allowed.
Scientific Calculators are allowed for calculations.

Section A (5 * 2= 10)

Q1. Two documents have:

- Same cosine similarity with query
- Different Euclidean distances

What does this indicate about: Direction vs magnitude?

Q2. When we are working with the ef_search parameter of HNSW algorithm why F1 score becomes important parameter along with accuracy to be considered?

Q3. Explain the type of Vector Based retriever we used which overcomes the differences in results due to poor embedding?

Q4. After updating embedding model, the retrieval quality drops. Explain: What should be updated along with embeddings?

Q5. You are given:

- Dataset size = 500,000 vectors
- Dimension = 256
- Query rate = 100 queries/sec

Explain:

- Which metric is better for high-dimensional embeddings.
- Justify using computation + behavior

Section B (10 * 2 = 20)

Q1.

a)

```python
def score(doc, weights):
    total = 0
    for i in range(len(doc)):
        total += doc[i] * weights[i]
    return total
```

Identify limitation of this scoring approach

b) You are given:

Term	D1	D2	D3
vector	2	1	0
retrieval	3	1	2
search	1	3	4

Query emphasis:

- vector → weight 0.2
- retrieval → weight 0.5
- search → weight 0.3

Explain:

- ✓ Why a document with lower total frequency can rank higher
- ✓ Which term dominates decision

Q2.

a)

```python
def search(node, query, graph):
    best = node
    for n in graph[node]:
        if abs(n - query) < abs(best - query):
            best = n
    return best
```

1. Why this may stop early
2. How to modify it to improve accuracy

b) You are given:
Query: Q = (2, -1, 2, 1)
Documents:
D1 = (4, -2, 4, 2)
D2 = (2, 0, 1, 3)
D3 = (1, -1, 2, 0)

- ✓ Compute cosine similarity: - Q vs each document.
- ✓ Why D1 may dominate in one metric but not the other